# Practice workbook · Read an enzyme optimum from data and judge how reliable it is

Tables, bar charts or graphs of enzyme activity against temperature or pH: reading or sketching the optimum, comparing enzymes, concluding, and judging how well the data pin down the optimum.

11 question parts: 1 worked-example retry and 10 further practice parts.
Name: _____ Class: _____ Date: _____

Practise one question type at a time. Attempt the part inside the red boundary; the rest of the question and any figures are kept for context. Write your working in the answer space.

The first question is a retry of the website worked example, without its solution. Check your answers online with 'Check my answer', or ask your teacher for the answer booklet.

OLevels.com SCHOOL

Type ID: HBI-ENZYME-OPTIMUM-FROM-DATA

# 1. Question 4 · part a.iii · 1 mark

June 2024 Paper 1R

Worked-example retry · Attempt part a.iii only.

**4** (a) The table gives information on how the time taken to digest starch by an enzyme varies at different pH levels.

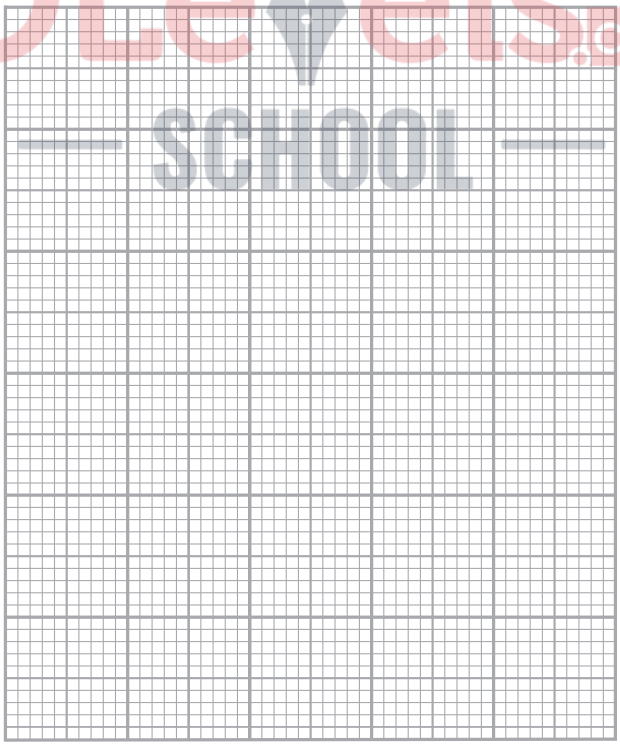
| pH | Time taken to digest starch in minutes |
|---|---|
| 2 | 60 |
| 4 | 45 |
| 6 | 20 |
| 8 | 35 |
| 10 | 55 |

(i) Plot a graph using the data in the table.

**(3)**

(ii) Draw a curve of best fit.

**(1)**



DO NOT WRITE IN THIS AREA
DO NOT WRITE IN THIS AREA
DO NOT WRITE IN THIS AREA

# 1. Question 4 · part a.iii · 1 mark

June 2024 Paper 1R

Worked-example retry · Attempt part a.iii only.

DO NOT WRITE IN THIS AREA

(iii) Use the graph to determine the optimum pH for the digestion of starch.

**(1)**

(iv) Describe the trend shown by the graph.

**(2)**

(v) Explain why the time taken to digest starch is different at pH 2 compared with the time taken at pH 6.

**(3)**

(vi) Name a part of the body where starch digestion takes place.

**(1)**

(b) Describe a test for starch in a sample of food.

**(2)**

**(Total for Question 4 = 13 marks)**

## 1. Question 4 · part a.iii · 1 mark · Working and answer

Working and answer

OLevels.com SCHOOL

# 2. Question 7 · part b.i · 1 mark

June 2019 Paper 1R

Further practice · Attempt part b.i only.

**7** Cheese is processed from milk. During this process, an enzyme is used to convert the lactose in milk into lactic acid.

A student uses this method to investigate the process.

- dissolve lactose in water to produce a solution
- put the lactose solution into three different test tubes
- keep the first test tube at 2 °C, the second test tube at 25 °C and the third test tube at 55 °C
- add enzyme solution to the lactose solution in each test tube
- measure the amount of lactic acid in the solution every 10 hours for a total of 60 hours

The table shows the student's results.

| Time after adding enzyme in hours | Amount of lactic acid in the solution in arbitrary units | | |
|---|---|---|---|
| | 2 °C | 25 °C | 55 °C |
| 0 | 0.1 | 0.1 | 0.1 |
| 10 | 0.1 | 0.2 | 0.2 |
| 20 | 0.1 | 0.3 | 0.2 |
| 30 | 0.2 | 0.8 | 0.4 |
| 40 | 0.3 | 1.8 | 0.7 |
| 50 | 0.4 | 2.2 | 1.0 |
| 60 | 0.6 | 2.3 | 1.3 |

(a) Describe what additional information another student would need to repeat this investigation.

**(4)**

DO NOT WRITE IN THIS AREA DO NOT WRITE IN THIS AREA DO NOT WRITE IN THIS AREA DO NOT WRITE IN THIS AREA

**20**

## 2. Question 7 · part b.i · 1 mark

June 2019 Paper 1R

Further practice · Attempt part b.i only.

(b) (i) Estimate the optimum temperature for this enzyme.

**(1)**

(ii) Explain why it is not possible to give an accurate value for the optimum temperature using the student's method.

**(2)**

(c) The student continued the investigation for a further 60 hours.

Explain the results that would be obtained.

**(3)**

DO NOT WRITE IN THIS AREA
DO NOT WRITE IN THIS AREA
DO NOT WRITE IN THIS AREA
DO NOT WRITE IN THIS AREA
DO NOT WRITE IN THIS AREA
DO NOT WRITE IN THIS AREA

OLevels.com SCHOOL

## 2. Question 7 · part b.i · 1 mark · Working and answer

Working and answer

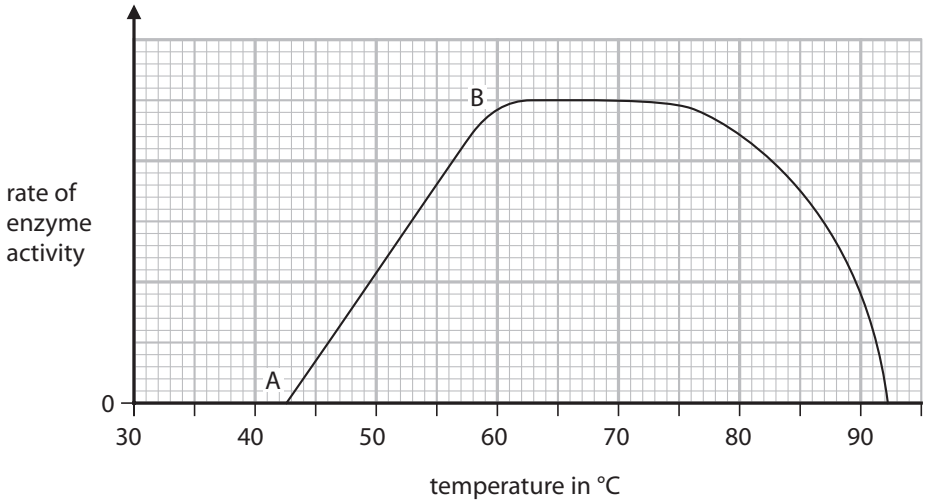
OLevels.com SCHOOL

## 3. Question 4 · part a · 2 marks

November 2020 Paper 2R

Further practice · Attempt part a only.

**4** The graph shows the effect of temperature on the activity of a bacterial enzyme.




(a) Draw a curve on the grid to show the rate of enzyme activity for an enzyme from the human alimentary canal.

**(2)**

(b) Explain what is causing the change in enzyme activity from A to B.

**(5)**

DO NOT WRITE IN THIS AREA
DO NOT WRITE IN THIS AREA
DO NOT WRITE IN THIS AREA

OLevels.com SCHOOL

## 3. Question 4 · part a · 2 marks · Working and answer

Working and answer

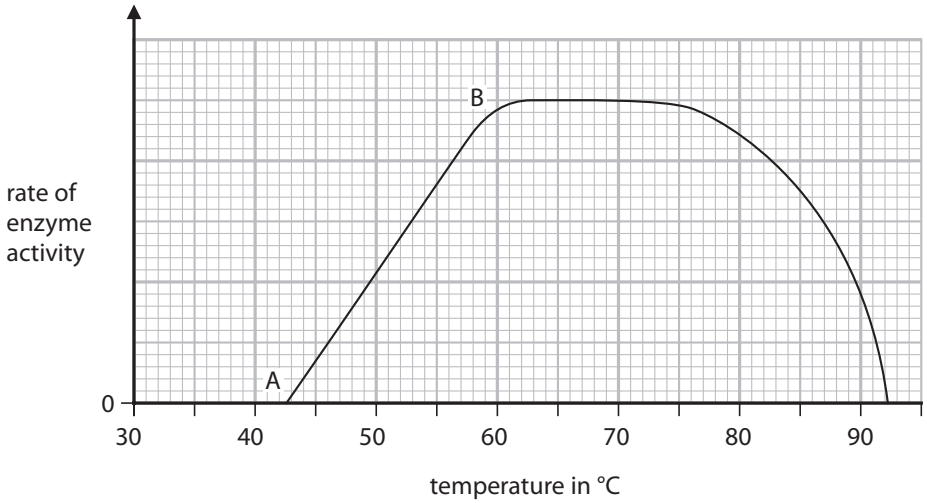
OLevels.com SCHOOL

# 4. Question 4 · part c · 2 marks

November 2020 Paper 2R

Further practice · Attempt part c only.

**4** The graph shows the effect of temperature on the activity of a bacterial enzyme.




(a) Draw a curve on the grid to show the rate of enzyme activity for an enzyme from the human alimentary canal.

**(2)**

(b) Explain what is causing the change in enzyme activity from A to B.

**(5)**

OLevels.com SCHOOL

DO NOT WRITE IN THIS AREA
DO NOT WRITE IN THIS AREA
DO NOT WRITE IN THIS AREA

# 4. Question 4 · part c · 2 marks

November 2020 Paper 2R

Further practice · Attempt part c only.

DO NOT WRITE IN THIS AREA

(c) Deduce which type of environment the bacterium lives in.

**(2)**

.....

.....

.....

.....

**(Total for Question 4 = 9 marks)**

Working and answer

OLevels.com SCHOOL

# 5. Question 3 · part b.iii · 3 marks

June 2021 Paper 1

Further practice · Attempt part b.iii only.

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

**3** Enzymes are protein molecules.

(a) (i) Which sub-units form enzymes?

**(1)**

- ☒ **A** amino acids
- ☒ **B** fatty acids
- ☒ **C** glucose
- ☒ **D** glycerol

(ii) Describe how a solution can be tested to see if it contains protein.

**(3)**

OLevels.com SCHOOL

# 5. Question 3 · part b.iii · 3 marks

June 2021 Paper 1

Further practice · Attempt part b.iii only.

(b) A scientist investigates the effect of temperature on two different enzymes, A and B.

Enzyme A and enzyme B both digest proteins.

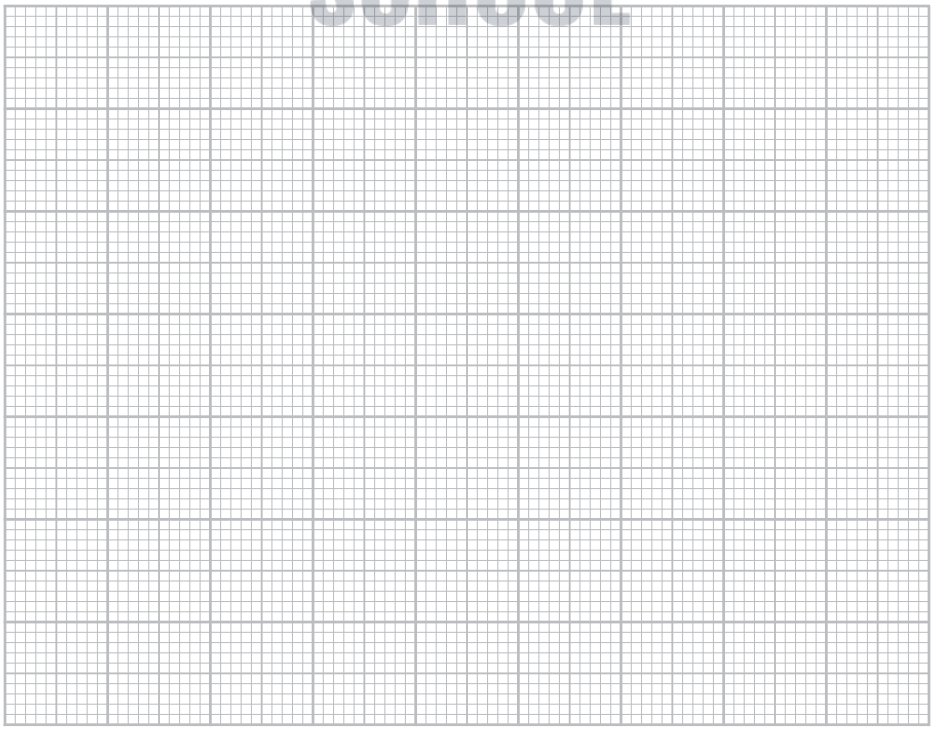
The results of the scientist's investigation are shown in the table.

| Temperature in °C | Rate of protein digestion in arbitrary units | |
|---|---|---|
| | Enzyme A | Enzyme B |
| 10 | 0 | 0 |
| 20 | 5 | 10 |
| 30 | 5 | 40 |
| 40 | 20 | 80 |
| 50 | 60 | 140 |
| 60 | 80 | 200 |
| 70 | 20 | 280 |
| 80 | 0 | 320 |
| 90 | 0 | 80 |

(i) Plot the scientist's results on the grid.

Draw the best fit curve for each enzyme.

**(5)**



DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA



P 6 6 4 3 2 A 0 8 2 4

# 5. Question 3 · part b.iii · 3 marks

June 2021 Paper 1

Further practice · Attempt part b.iii only.

DO NOT WRITE IN THIS AREA

(ii) Describe the effects on enzyme B of increasing the temperature from 80°C to 90°C.

**(3)**

DO NOT WRITE IN THIS AREA

(iii) Describe the differences between the rate of protein digestion shown by enzyme A and enzyme B.

Refer to the graph in your answer.

**(3)**

**(Total for Question 3 = 15 marks)**

OLevels.COM — SCHOOL —

## 5. Question 3 · part b.iii · 3 marks · Working and answer

Working and answer

OLevels.com SCHOOL

# 6. Question 1 · part b.i · 1 mark

November 2021 Paper 1

Further practice · Attempt part b.i only.

**1** The enzyme catalase is found in the liver.

Catalase acts as a catalyst when added to hydrogen peroxide solution, causing the solution to break down into water and oxygen.

A student investigates the increase in the rate of breakdown of hydrogen peroxide into water and oxygen using catalase.

This is the student's method.

- put 100 g of liver into 100 $cm^3$ of water, and mix into a smooth paste
- put 10 $cm^3$ of the paste into a beaker at room temperature
- add one drop of hydrogen peroxide to the paste
- assess how many oxygen bubbles are produced on a scale of 0 to 5, with 5 representing most bubbles and 0 representing no bubbles

The student repeats the investigation using different conditions.

The conditions he uses are:

- liver paste heated to 70 °C
- liver paste that has been kept at 3 °C for an hour
- acidified liver paste
- liver paste with an alkali added
- liver paste with no hydrogen peroxide added

These are the student's results.

*room temperature = 5  heated to 70 °C = 0*

*kept at 3 °C = 1  acid added = 2  alkali added = 3*

*no hydrogen peroxide added = 0*

DO NOT WRITE IN THIS AREA
DO NOT WRITE IN THIS AREA
DO NOT WRITE IN THIS AREA

# 6. Question 1 · part b.i · 1 mark

November 2021 Paper 1

Further practice · Attempt part b.i only.

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

(a) (i) Give the student's results in the form of a suitable table.

**(3)**

(ii) State why this investigation is not reliable.

**(1)**

OLevels.COM — SCHOOL —

(iii) Give a reason why the results of this investigation are not accurate.

**(1)**

(iv) Describe how the investigation could be improved so that the conclusions could be more accurate and reliable.

**(3)**

## 6. Question 1 · part b.i · 1 mark

November 2021 Paper 1

Further practice · Attempt part b.i only.

(b) (i) State the condition where the enzyme was most active.

**(1)**

(ii) State the reason for including the beaker with no hydrogen peroxide added.

**(1)**

(iii) Explain the result obtained when the conditions were acidic.

**(4)**

**(Total for Question 1 = 14 marks)**

DO NOT WRITE IN THIS AREA
DO NOT WRITE IN THIS AREA

Working and answer

# 7. Question 7 · part a.ii · 1 mark

November 2023 Paper 2

Further practice · Attempt part a.ii only.

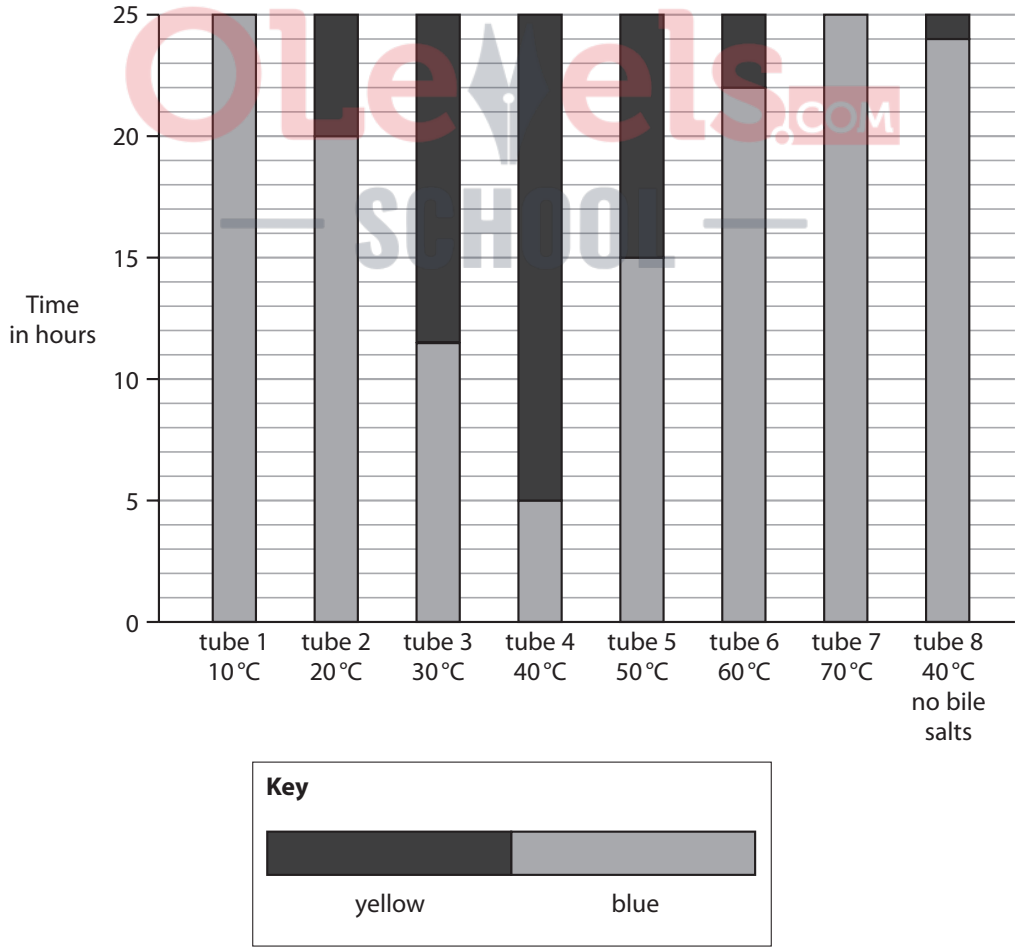
**7** A student investigates the digestion of olive oil (lipid) using the enzyme lipase, bile salts and a blue indicator.

The indicator changes from blue to yellow when the pH is below 5.

This is the student's method.

- mix together olive oil, lipase solution, bile salts and one drop of indicator
- put this mixture into a test tube
- repeat the process for six other test tubes
- keep the tubes at different temperatures
- keep another test tube (test tube 8) containing just olive oil, indicator and lipase at 40 °C
- record the colours of the mixture of each tube every hour for 25 hours

The chart shows the student's results.




DO NOT WRITE IN THIS AREA
DO NOT WRITE IN THIS AREA
DO NOT WRITE IN THIS AREA

# 7. Question 7 · part a.ii · 1 mark

November 2023 Paper 2

Further practice · Attempt part a.ii only.

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

(a) (i) State three factors that should be kept constant in the student's investigation.

**(3)**

1 .....

2 .....

3 .....

(ii) State the optimum temperature for the action of lipase.

**(1)**

.....

(iii) Discuss why this temperature is only an approximate optimum temperature.

**(3)**

.....

.....

.....

.....

.....

.....

OLevels.com SCHOOL

(b) (i) Determine how much more time the indicator takes to change colour in tube 8 than in tube 3.

**(2)**

time = ..... hours

(ii) Explain why the pH decreases during the 25 hours.

**(3)**

.....

.....

.....

.....

.....

.....

## 7. Question 7 · part a.ii · 1 mark · Working and answer

Working and answer

OLevels.com SCHOOL

## 8. Question 7 · part e · 2 marks

November 2024 Paper 2

Further practice · Attempt part e only.

**7** Six students investigate the activity of an enzyme found in the alimentary canal.

This enzyme digests a cloudy suspension into a clear solution.

The results of the investigation are shown in the table.

| Student | Volume of enzyme solution in $cm^3$ | Temperature in °C | pH | Volume of cloudy suspension in $cm^3$ | Time tubes left in minutes | Appearance at end of investigation |
|---|---|---|---|---|---|---|
| 1 | 5 | 37 | 2 | 10 | 9 | cloudy |
| 2 | 5 | 37 | 2 | 10 | 20 | clear |
| 3 | 5 | 37 | 4 | 10 | 35 | clear |
| 4 | 5 | 37 | 4 | 10 | 20 | cloudy |
| 5 | 5 | 37 | 6 | 10 | 8 | cloudy |
| 6 | 5 | 37 | 6 | 10 | 40 | cloudy |

(a) Explain why the investigation was carried out at 37 °C.

**(2)**

(b) Explain which property of the enzyme the students are investigating.

**(2)**

DO NOT WRITE IN THIS AREA DO NOT WRITE IN THIS AREA DO NOT WRITE IN THIS AREA

## 8. Question 7 · part e · 2 marks

November 2024 Paper 2

Further practice · Attempt part e only.

DO NOT WRITE IN THIS AREA
DO NOT WRITE IN THIS AREA
DO NOT WRITE IN THIS AREA

(c) Comment on the results of the investigation.

In your answer include ideas of validity and accuracy.

**(4)**

OLevels.com SCHOOL

(d) Describe a control experiment that the students could have carried out.

**(3)**

## 8. Question 7 · part e · 2 marks

November 2024 Paper 2

Further practice · Attempt part e only.

(e) State a possible conclusion that the students could make from the results of this investigation.

**(2)**

(f) Explain where this enzyme is likely to be found in the alimentary canal.

**(2)**

**(Total for Question 7 = 15 marks)**

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS ARI

OLevels.com SCHOOL

Working and answer

# 9. Question 7 · part d · 5 marks

June 2019 Paper 1R

Further practice · Attempt part d only.

**7** Cheese is processed from milk. During this process, an enzyme is used to convert the lactose in milk into lactic acid.

A student uses this method to investigate the process.

- dissolve lactose in water to produce a solution
- put the lactose solution into three different test tubes
- keep the first test tube at 2 °C, the second test tube at 25 °C and the third test tube at 55 °C
- add enzyme solution to the lactose solution in each test tube
- measure the amount of lactic acid in the solution every 10 hours for a total of 60 hours

The table shows the student's results.

| Time after adding enzyme in hours | Amount of lactic acid in the solution in arbitrary units | | |
|---|---|---|---|
| | 2 °C | 25 °C | 55 °C |
| 0 | 0.1 | 0.1 | 0.1 |
| 10 | 0.1 | 0.2 | 0.2 |
| 20 | 0.1 | 0.3 | 0.2 |
| 30 | 0.2 | 0.8 | 0.4 |
| 40 | 0.3 | 1.8 | 0.7 |
| 50 | 0.4 | 2.2 | 1.0 |
| 60 | 0.6 | 2.3 | 1.3 |

(a) Describe what additional information another student would need to repeat this investigation.

**(4)**

DO NOT WRITE IN THIS AREA DO NOT WRITE IN THIS AREA DO NOT WRITE IN THIS AREA DO NOT WRITE IN THIS AREA

**20**

## 9. Question 7 · part d · 5 marks

June 2019 Paper 1R

Further practice · Attempt part d only.

(b) (i) Estimate the optimum temperature for this enzyme.

**(1)**

(ii) Explain why it is not possible to give an accurate value for the optimum temperature using the student's method.

**(2)**

(c) The student continued the investigation for a further 60 hours.

Explain the results that would be obtained.

**(3)**

DO NOT WRITE IN THIS AREA
DO NOT WRITE IN THIS AREA
DO NOT WRITE IN THIS AREA
DO NOT WRITE IN THIS AREA
DO NOT WRITE IN THIS AREA
DO NOT WRITE IN THIS AREA

OLevels.com SCHOOL

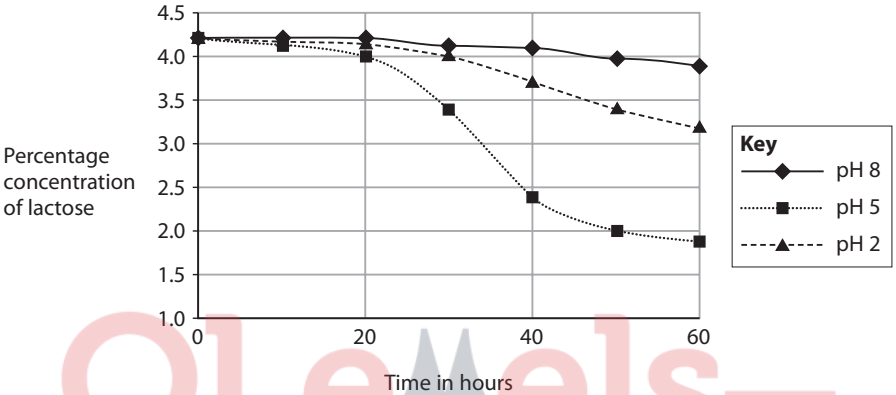
# 9. Question 7 · part d · 5 marks

June 2019 Paper 1R

Further practice · Attempt part d only.

(d) The student uses the same enzyme to investigate the effect of pH on the conversion of the lactose into lactic acid.

The student's results are shown in the graph.




The student wants to determine the most effective conditions for the conversion of lactose into lactic acid in cheese manufacture.

OLevels.com SCHOOL

DO NOT WRITE IN THIS AREA
DO NOT WRITE IN THIS AREA
DO NOT WRITE IN THIS AREA

## 9. Question 7 · part d · 5 marks

June 2019 Paper 1R

Further practice · Attempt part d only.

Discuss if this is possible using the results of the two investigations.

**(5)**

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

OLevels.com SCHOOL

**(Total for Question 7 = 15 marks)**

## 9. Question 7 · part d · 5 marks · Working and answer

Working and answer

OLevels.com SCHOOL

# 10. Question 7 · part a.iii · 3 marks

November 2023 Paper 2

Further practice · Attempt part a.iii only.

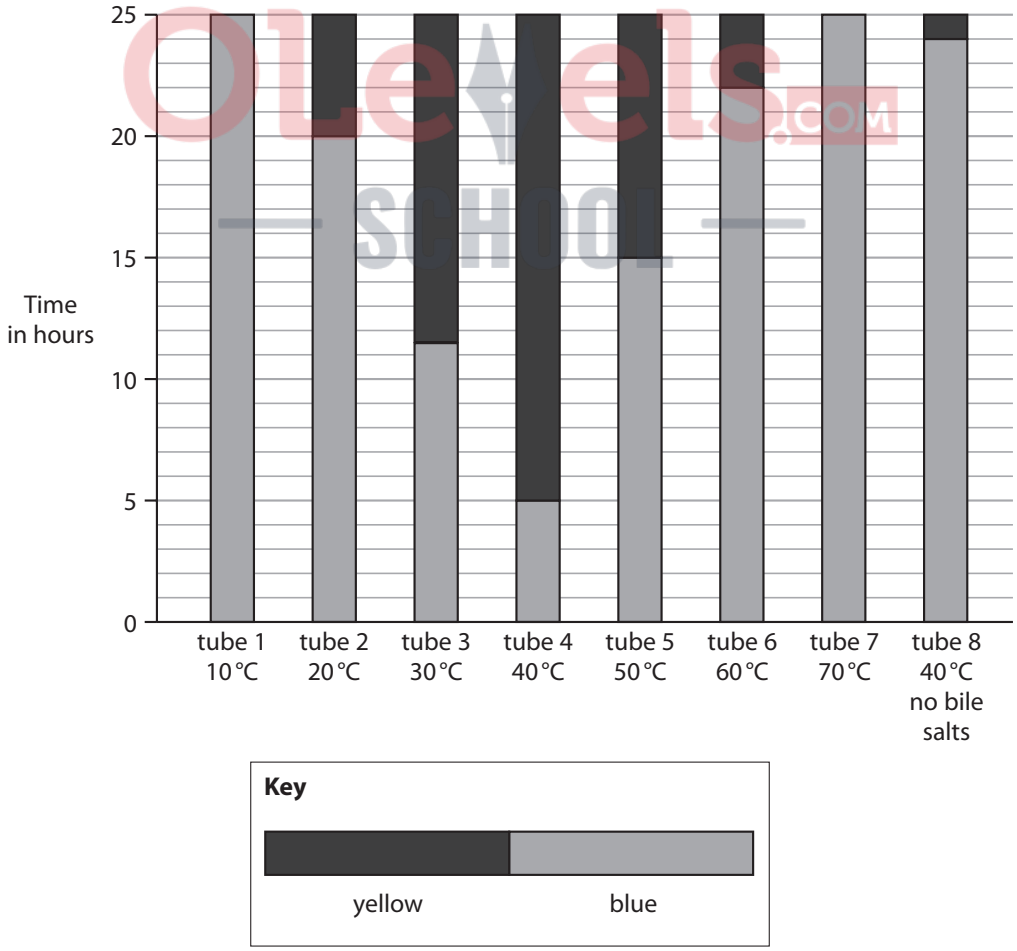
**7** A student investigates the digestion of olive oil (lipid) using the enzyme lipase, bile salts and a blue indicator.

The indicator changes from blue to yellow when the pH is below 5.

This is the student's method.

- mix together olive oil, lipase solution, bile salts and one drop of indicator
- put this mixture into a test tube
- repeat the process for six other test tubes
- keep the tubes at different temperatures
- keep another test tube (test tube 8) containing just olive oil, indicator and lipase at 40 °C
- record the colours of the mixture of each tube every hour for 25 hours

The chart shows the student's results.




DO NOT WRITE IN THIS AREA
DO NOT WRITE IN THIS AREA
DO NOT WRITE IN THIS AREA

## 10. Question 7 · part a.iii · 3 marks

November 2023 Paper 2

Further practice · Attempt part a.iii only.

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

(a) (i) State three factors that should be kept constant in the student's investigation.

**(3)**

1 .....

2 .....

3 .....

(ii) State the optimum temperature for the action of lipase.

**(1)**

.....

(iii) Discuss why this temperature is only an approximate optimum temperature.

**(3)**

.....

.....

.....

.....

.....

.....

OLevels.com SCHOOL

(b) (i) Determine how much more time the indicator takes to change colour in tube 8 than in tube 3.

**(2)**

time = ..... hours

(ii) Explain why the pH decreases during the 25 hours.

**(3)**

.....

.....

.....

.....

.....

.....

## 10. Question 7 · part a.iii · 3 marks · Working and answer

Working and answer

OLevels.com SCHOOL

## 11. Question 7 · part c · 4 marks

November 2024 Paper 2

Further practice · Attempt part c only.

**7** Six students investigate the activity of an enzyme found in the alimentary canal.

This enzyme digests a cloudy suspension into a clear solution.

The results of the investigation are shown in the table.

| Student | Volume of enzyme solution in $cm^3$ | Temperature in °C | pH | Volume of cloudy suspension in $cm^3$ | Time tubes left in minutes | Appearance at end of investigation |
|---|---|---|---|---|---|---|
| 1 | 5 | 37 | 2 | 10 | 9 | cloudy |
| 2 | 5 | 37 | 2 | 10 | 20 | clear |
| 3 | 5 | 37 | 4 | 10 | 35 | clear |
| 4 | 5 | 37 | 4 | 10 | 20 | cloudy |
| 5 | 5 | 37 | 6 | 10 | 8 | cloudy |
| 6 | 5 | 37 | 6 | 10 | 40 | cloudy |

(a) Explain why the investigation was carried out at 37 °C.

**(2)**

(b) Explain which property of the enzyme the students are investigating.

**(2)**

DO NOT WRITE IN THIS AREA DO NOT WRITE IN THIS AREA DO NOT WRITE IN THIS AREA



P 7 5 9 5 9 A 0 2 2 2 4

## 11. Question 7 · part c · 4 marks

November 2024 Paper 2

Further practice · Attempt part c only.

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

(c) Comment on the results of the investigation.

In your answer include ideas of validity and accuracy.

**(4)**

OLevels.com SCHOOL

(d) Describe a control experiment that the students could have carried out.

**(3)**

## 11. Question 7 · part c · 4 marks · Working and answer

Working and answer

OLevels.com SCHOOL